

Yang Ni

✉ ny22@mails.tsinghua.edu.cn 🌐 yni-astro.github.io

🎓 Education

Tsinghua University Ph.D. Candidate in Astronomy, Advisor: Prof. Xuening Bai	2022 – Present <i>Beijing, China</i>
Princeton University Visiting Student Research Collaborators (VSRC), Host Advisor: Prof. Jeremy Goodman	2026.02 – 2026.08 <i>New Jersey, US</i>
Nanjing University B.S. in Astronomy (Magna Cum Laude), Advisor: Prof. Bin Luo	2018 – 2022 <i>Nanjing, China</i>
Massachusetts Institute of Technology Summer Research (remote), Host Advisor: Prof. Mark Vogelsberger	2020 – 2021 <i>Cambridge, US</i>

🔬 Research Interests

- Early-stage evolution of protoplanetary disks; planet formation
- Star formation in AGN disks
- Evolution of giant molecular clouds; star formation in ISM

📄 Publications

(**F** for first author, **C** for co-author)

F4. *Radiation Hydrodynamics of Self-gravitating Protoplanetary Disks II. Accretion, Migration, Spin, and Internal Structure of GI Fragments,*

Y. Ni, H. Deng, X. Bai To be submitted, arXiv:2609.37984

F3. *Modeling the Evolution of Protoplanetary Disks: Two Pathways from Gravitational Instability to MHD Wind-Driven Accretion,*

Y. Ni, W. Xu, X. Bai re-submitted to ApJ (minor revision), arXiv:2607.12007

F2. *Radiation Hydrodynamics of Self-gravitating Protoplanetary Disks I. Direct Formation of Gas Giants via Disk Fragmentation,*

Y. Ni, H. Deng, X. Bai ApJ, 995, 96

F1. *The life cycle of giant molecular clouds in simulated Milky Way-mass galaxies,*

Y. Ni, H. Li, M. Vogelsberger, L. Sales, F. Marinacci, P. Torrey A&A, 699, A282

C3. *RIGEL: Feedback regulated cloud-scale star formation efficiency in a simulated dwarf galaxy merger, Y. Deng, ..., **Y. Ni**, et al.* A&A, 704, A240

C2. *The THESAN-ZOOM project: Star formation efficiency from giant molecular clouds to galactic scale in high-redshift starbursts,*

Z. Wang, ..., **Y. Ni**, et al. MNRAS, 544, 2675

C1. *GRB 210121A: A Typical Fireball Burst Detected by Two Small Missions,*

X. I. Wang, ..., **Y. Ni**, et al. ApJ, 922, 237

👤 Talks and Posters

Selected:

Protoplanetary Disks & Exoplanets 2026, The Kavli Institute for Astronomy and Astrophysics (KIAA) at Peking University, China	September 2026
(Invited talk) Purple Mountain Observatory, Jiangsu, China	August 2026
AthenaK Summer School, Penn State University, US	July 2026
Exoplanets and Planet Formation Mini-Workshop, Tsinghua University, China	December 2025
ICEPF Conference 2025, Shanghai, China (<i>Poster</i>)	December 2025
East Asia Numerical Astrophysics Meeting (EANAM10), Jeju, Korea	September 2025
Exoplanets & Planet Formation Workshop 2025, Yunnan, China	August 2025
NSFC Scientific Meeting on CPDs and Satellite Formation, Beijing, China	May 2024
Exoplanets & Planet Formation Workshop 2023, Beijing, China (<i>Poster</i>)	December 2023
International Conference on Resolving Galaxy Ecosystems Across All Scales, The Chinese University of Hong Kong, China	December 2023
Colloquium, Department of Astrophysics, Universität Zürich, Switzerland	November 2023

Technical Skills

- ✔ Contributor and experienced user of the multi-grid self-gravity module in AthenaK
- ✔ Contributor and experienced user of the M1-radiative transfer module in GIZMO (PLM reconstruction, RT subcycling, implicit source solver)
- ✔ Experienced user of the radiative (magneto-)hydrodynamic simulation with GIZMO
- ✔ Experienced User of IRAF/Pyraf; The Chandra Interactive Analysis of Observations (CIAO)
- ✔ Experienced User of Python: astropy, h5py, numpy, matplotlib, scipy, yt, networkx... ; C/C++; Mathematica; L^AT_EX
- ✔ Experienced User of AI agent harness including Claude Code and Codex
- ✔ Experienced User of LLMs including Claude Opus 4.6/4.7/4.8/5, Claude Fable 5, GPT 5.5/5.6/6, Gemini 3.1 Pro/3.5 Flash

Honors & Awards

Selected:

Tsinghua University Merit Student,	Tsinghua University
Tsinghua Scholarship for Overseas Graduate Studies,	Tsinghua University
Comprehensive Scholarship,	Tsinghua University
National Astronomical Observatory of China (NAOC) Scholarship,	NAOC
People's Scholarship First Prize,	The Ministry of Education of the PRC
Outstanding Undergraduate,	Nanjing University
"Top-notch" First Prize,	Nanjing University
G. Zheng Scholarship for Oversea Study,	Zheng Foundation

References

(PhD Advisor) **Prof. Xuening Bai**, Tsinghua University

✉ xbai@tsinghua.edu.cn

(PhD Co-Advisor) **Prof. Hongping Deng**, Shanghai Astronomical Observatory (SHAO)

✉ hpdeng353@shao.ac.cn

Prof. Jeremy Goodman, Princeton University

✉ jeremy@astro.princeton.edu

Prof. Mark Vogelsberger, Massachusetts Institute of Technology

✉ mvogelsb@mit.edu

Prof. Hui Li, Tsinghua University

✉ hliastro@tsinghua.edu.cn

Prof. Bin Luo, Nanjing University

✉ bluo@nju.edu.cn